



Holography

PHYS3114

Student Notes

1 Experimental Aim

This experiment is designed to give some practical experience in recording and viewing holographic images and to understand these processes in the context of diffraction theory and Fourier optics.

1.1 Learning Outcomes

- Explain the physical principles behind holography, how it is distinguished from conventional image recording, and the requirements for obtaining a hologram.
- Use scalar diffraction theory to give a basic mathematical description of the role of the reference beam in recording / viewing transmission holograms.
- Calculate the optimal exposure time recording a hologram for a given optical arrangement and recording medium.
- Use Fourier theory to explain how changes in the illumination conditions affect the observed images from a transmission hologram.
- Provide a physical interpretation of the difference in the recording method between thick and thin holograms and explain how it impacts the viewing conditions.
- Explain how double exposure holographic interferometry works and why they are useful.
- Identify some modern applications of holography and discuss why they are used over a competing technology.

2 Introduction

In 1948, Hungarian physicist Dennis Gabor devised a method for faithfully recording all phase and amplitude information of an optical field using a recording media which

only responds to light intensity. Gabor's wavefront reconstruction method is based on mixing the light from an object with a plane wave reference beam to create a complex interference pattern that is captured by a suitable photosensitive recording medium. The interference patterns then act as a series of diffraction gratings that, when illuminated by the reference beam in the absence of the object, reproduce the object field. This is the basis of holography and Gabor was awarded the 1971 Nobel Prize for his discovery. As holography makes use of coherent light, many advances in holography occurred after the invention of the laser in 1961.



Figure 1: Help me Obi-Wan Kenobi, this technology doesn't have a hope!

Advances in recording media and coherent light sources in the 1970s and 80s allowed researchers to explore the capabilities of holography as a display technology, as well as novel applications, such as holographic data storage. Photorefractive crystals were considered for holographic video. Holography as an art medium was also explored over the period, and whilst it never really made mainstream popularity, it did offer some rather unique ways to explore visual effects. In fact, for many years UNSW College of Fine Arts (now the School of Art and Design) was home to Australia's preeminent holographer, Paula Dawson AM. The optical table we use for our holography work was kindly donated to the School of Physics Teaching Lab by Paula in 2023.

With advances in computational processing powers through the 1990s and the emergence of spatial light modulator (SLM) technologies, digital holographic techniques became a prominent area of research. Typical problems for this area would involve calculating phase patterns that, when applied to light fields, would generate target intensity profile under far-field diffraction conditions. This is not as trivial as it sounds because the information about the final intensity patterns doesn't contain any direct phase information, and so iterative algorithms need to be applied. Another example of digital holography is to reconstruct images based on optical techniques that retrieve phase information; this is the basic premise behind holographic microscopy.

2.1 A taxonomy of holograms

Holograms are classified according to:

- the thickness of the recording material
- the type of modulation imposed on the illuminating (readout wave)
- the recording geometry

- the mode of image formation

A **thin** (plane) hologram is one where the thickness of the recording medium is smaller than the fringe spacing. In **thick** (volume) holograms the thickness is greater than the fringe spacing.

With **amplitude** holograms the interference pattern is recorded as an optical density variation of the medium, which leads to absorption (amplitude modulation) of part of the illuminating wave. In **phase** holograms the refractive index of the recording medium varies with position, imposing a phase modulation onto the illuminating beam.

A **Fresnel** hologram is obtained when the object is sufficiently close to the recording medium so that the Fresnel diffraction pattern of the object is recorded. If the distance is large enough to record the far field pattern, a **Fraunhofer** hologram results. A **Fourier transform** hologram is produced when a lens, one focal length distant from both the object and the recording medium, is used to give the far field pattern of the object. In this last case the reference beam must be a plane wave. If the reference beam originates from a point the same distance from the recording medium as the object, **lensless Fourier transform** hologram is formed.

An **in-line** hologram is obtained when the two beams, (object and reference) interfering at the recording medium, are travelling in nearly the same direction. If their directions are substantially different, an off-axis hologram results while, if the two beams are travelling in essentially opposite directions a **reflection** hologram is formed. The image, in the case of off-axis and in-line hologram, is formed by passing a readout beam through the hologram, so these are known as **transmission** holograms. In the case of reflection holograms, an image is formed when the readout is **reflected** from the hologram.

3 Theory

3.1 Conventional Recording

Consider a monochromatic plane wave of the form, $E(\mathbf{r}, t) = E_0 \cos(\mathbf{k} \cdot \mathbf{r} - \omega t)$, which is a solution of Maxwell's Eqn. in the absence of charges and currents. At point, $\mathbf{r}$, the instantaneous intensity, $I(\mathbf{r}, t)$, is given by the squared magnitude of the complex phasor,

$$I(\mathbf{r}, t) = E_0^2 \cos^2(\mathbf{k} \cdot \mathbf{r} - \omega t) = \frac{E_0^2}{2} (1 - \cos(2(\mathbf{k} \cdot \mathbf{r} - \omega t))) \quad (1)$$

Almost all light measuring instruments that you can think of (... photodetectors, CCD cameras, photographic film, your retina ...) are sensitive only to light intensity or irradiance. Moreover, as the temporal oscillations in the optical region of the electromagnetic spectrum are so fast ($\approx 10^{15}$ Hz) compared to the response time of the medium, only the time averaged intensity, $\langle I \rangle_t$, is recorded. This essentially means that the time harmonic component will average out to zero leaving only the constant amplitude.

3.2 Recording the total field

To analyse problems in holography (and diffraction problems in general) we can use the language of scalar optical wave functions. This allows us to keep track of phases and amplitudes of multiple interfering fields without needing to invoke the full Maxwell treatment. Optical wave functions are solutions to the scalar wave equation and usually expressed in terms of complex plane waves,

$$U(r, t) = Ae^{i(kr - \omega t)} \quad (2)$$

The corresponding intensity of the optical wave function is then,

$$I(r, t) = U(r, t)U(r, t)^* = |A|^2. \quad (3)$$

Now, consider the interference between two coherent waves incident on the recording medium, as depicted in Figure 2. There object beam can be expressed as,

$$a(x, y) = |a(x, y)|e^{i\phi(x, y)}, \quad (4)$$

and the reference beam,

$$A(x, y) = |A(x, y)|e^{i\psi(x, y)}. \quad (5)$$

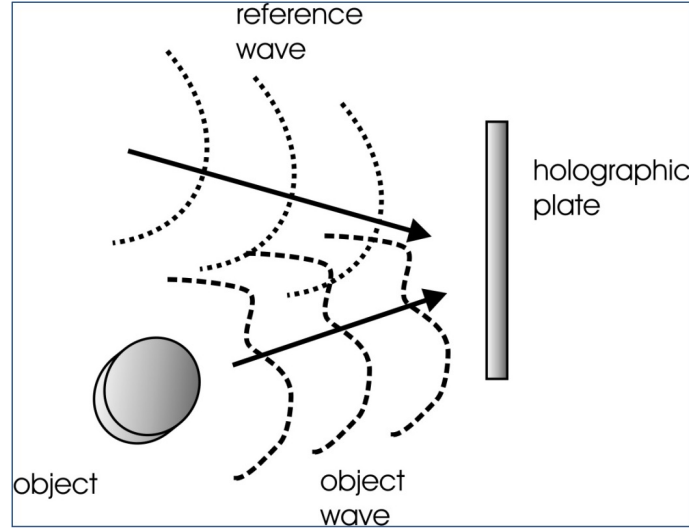


Figure 2: Idealised recording configuration including a field.

The total intensity on the plate is given by,

$$\begin{aligned} I(x, y) &= (a(x, y) + A(x, y)) (a(x, y) + A(x, y))^* \\ &= (|a(x, y)|e^{i\phi(x, y)} + |A(x, y)|e^{i\psi(x, y)}) (|a(x, y)|e^{-i\phi(x, y)} + |A(x, y)|e^{-i\psi(x, y)}) \\ &= |A(x, y)|^2 + |a(x, y)|^2 + 2|A(x, y)||a(x, y)| \cos(\phi(x, y) - \psi(x, y)) \end{aligned} \quad (6)$$

You can see that phase information now appears in the intensity profile, which is one of the key requirements for capturing all information about the field. For those familiar with signal processing, this is a little like an optical analogue of homodyne detection. We now make a couple of assumptions. Firstly, the recording medium responds linearly with intensity. Secondly, the recording process modulates the transmission of the film. In practice the films we use modulate the refractive index of the film, which encodes the intensity pattern as a phase response, but let's assume amplitude hologram in the first instance. Finally, we assume that reference beam is reasonably homogenous across the face of the hologram.

The recorded film transmission now becomes:

$$t_A(x, y) = t_b + \beta' \left[|a(x, y)|^2 + A(x, y)^* a(x, y) + A(x, y) a(x, y)^* \right] \quad (7)$$

where the first term is the background transmittance from the reference beam, the second is the intensity produced solely by the object beam, and the other terms relate to the interference between two fields. The constant β' accounts for the relative response of the recording medium.

3.3 Recovering the phase information

The holographic recording is now illuminated by a coherent reconstruction beam, $B(x, y)$. The transmitted field then becomes,

$$B(x, y)t_A(x, y) = t_b B + \beta' a a^* B + \beta' A^* a B + \beta' A a^* B. \quad (8)$$

In the case where B is just equal to the original reference wave, A , the third term on the right hand side is proportional to the original object wave:

$$\beta' |A|^2 a(x, y) \quad (9)$$

Alternatively, if the reconstruction beam is the conjugate of the reference beam, then the fourth term,

$$\beta' |A|^2 a(x, y)^*, \quad (10)$$

contains the conjugate of the original object wave. The physical implications of the having a conjugate beam is that the wavefront is inverted such that points on the leading edge become lagging ones. Phase conjugation is an interesting optical phenomenon that has many potential applications, but it is difficult to achieve in practice.

The original version of the hologram recording configuration proposed by Gabor was that a single beam was used as both the reference and object beam. It meant that the objects were necessarily weakly scattering and also that all images (real, virtual, etc.) were simultaneously observable.

3.4 The transmission hologram

To circumvent the problems of the Gabor hologram an alternative recording geometry was proposed where the reference beam and object beam were incident on the holographic

plate at different angles. The optical field incident on the recording medium is, in this case, given by,

$$U(x, y) = A \exp\left(-i2\pi y \frac{\sin(2\theta)}{\lambda}\right) + a(x, y), \quad (11)$$

and the intensity is given by,

$$I(x, y) = \left(|A| \exp\left(-i2\pi y \frac{\sin(2\theta)}{\lambda}\right) + a(x, y) \right) \left(|A| \exp\left(+i2\pi y \frac{\sin(2\theta)}{\lambda}\right) + a(x, y)^* \right), \quad (12)$$

Upon reading out the hologram with the reconstruction beam, B, the real and virtual images are spatially separated from each other and the directly transmitted beam:

$$\begin{aligned} & \beta' B A^* a(x, y) \exp\left(i2\pi y \frac{\sin(2\theta)}{\lambda}\right) \\ & \beta' B A a(x, y)^* \exp\left(-i2\pi y \frac{\sin(2\theta)}{\lambda}\right) \end{aligned} \quad (13)$$

Note that there is a minimum requirement for the angle between object and reference beam in order to clearly separate all component.

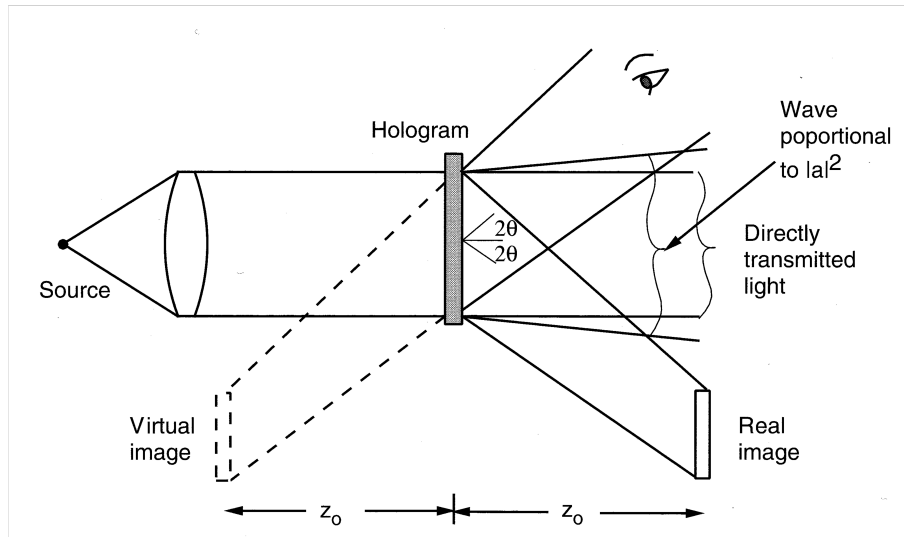


Figure 3: Reading out an off-axis transmission hologram. Taken from Goodman, Fourier Optics.

One of the major drawbacks of the transmission holograms is that you can't read it out using a white light source. The position where the image is formed depends on the wavelength, similar to the way a diffraction grating disperses light. The magnification of the image is also linked to the readout wavelength. This leads to a blurred rainbow image where the blue images appear smaller than the red ones. The more polychromatic the light source the worse the effect (i.e. a LED is better than a halogen lamp).